

Car Price Analysis Dashboard

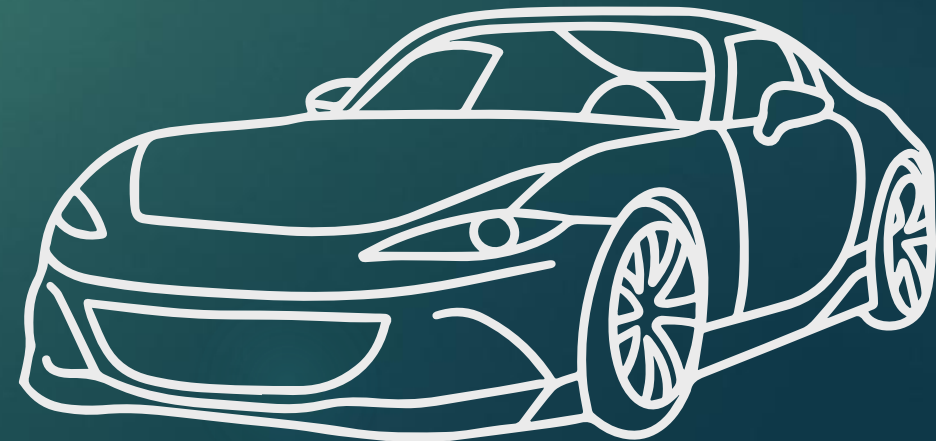
Hackathon Project by Team NaMiPaJay Crew

Ajay Chohan (Project Manager)

Nasra Ibrahim (Data Architect)

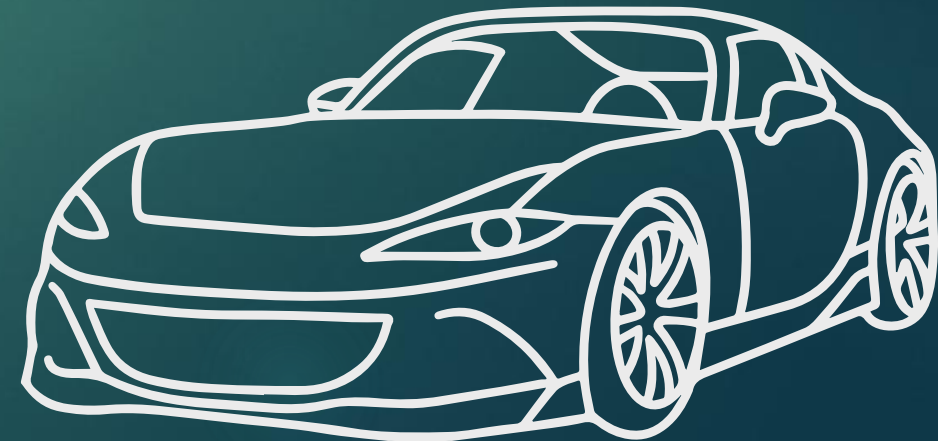
Milos Visnjic (Data Architect)

Payal Kanani (Data Analyst)



Problem Statement

- ▶ We want to understand what affects car prices in the used car market.
- ▶ Which factors—like brand, fuel type, horsepower, or engine type—impact price the most?
- ▶ This analysis will help support marketing, pricing strategy, and product development.

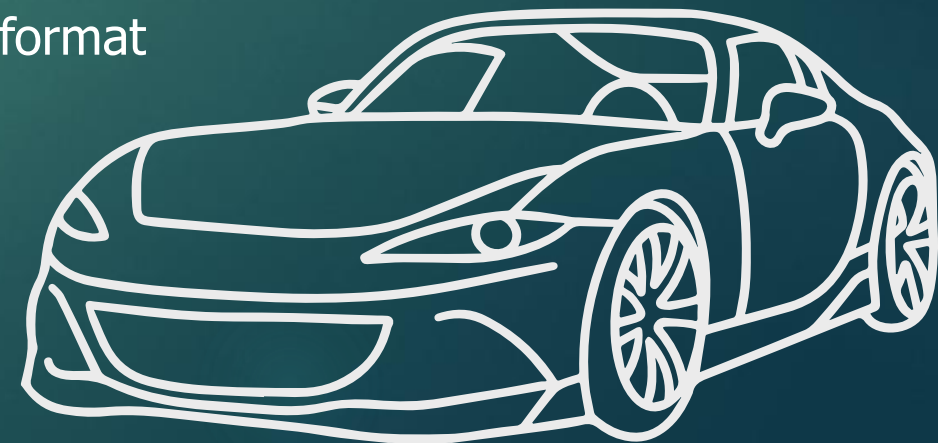


Data Overview

- ▶ Dataset: Cleaned version of the Kaggle Car Price Prediction dataset.
- ▶ Source: <https://www.kaggle.com/datasets/hellbuoy/car-price-prediction>
- ▶ Key Features: Car Name, Brand, Fuel Type, Mileage, Engine, Power, Seats, Price, Engine type, etc.

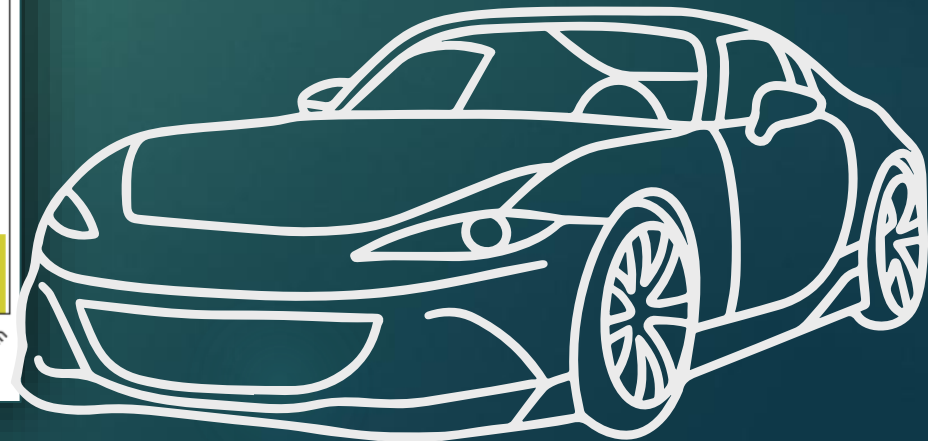
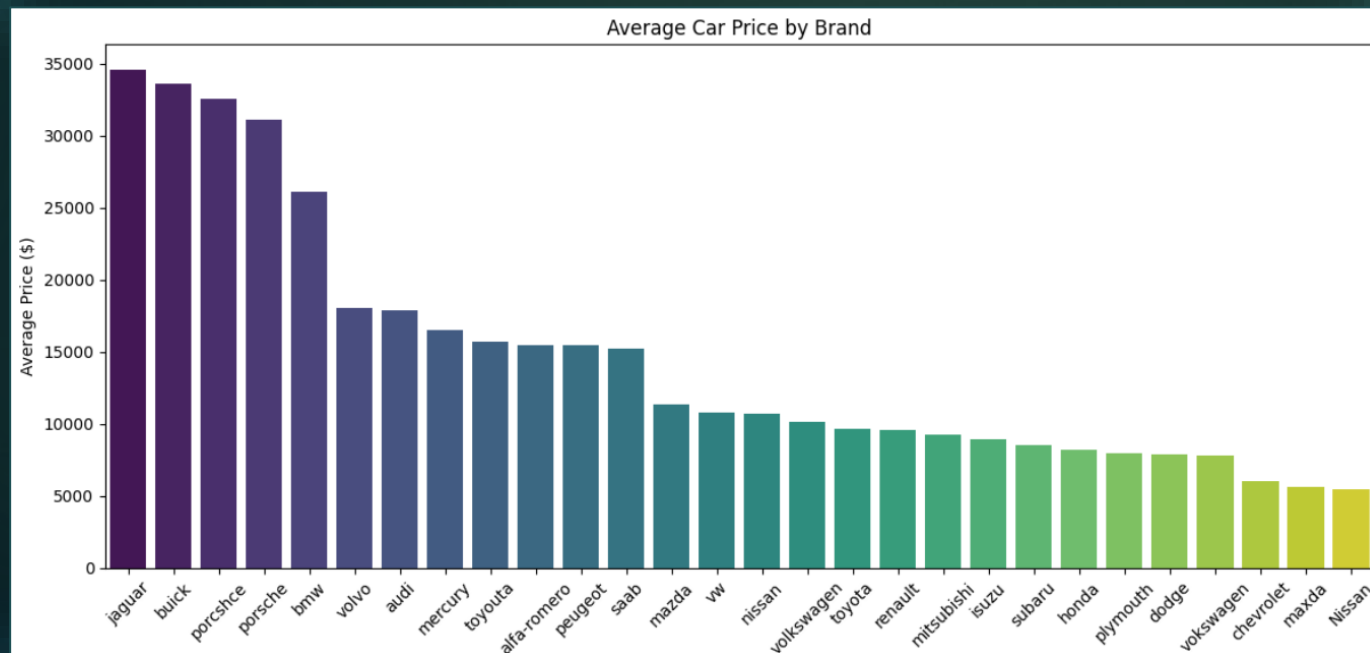
Data Cleaning

- ▶ Removed missing and null values (e.g., empty power or mileage)
- ▶ Converted text fields to categorical values (e.g., Fuel Type, Brand)
- ▶ Fixed inconsistent values in columns (e.g., 'VW' → 'Volkswagen')
- ▶ Split 'Car Name' into 'Brand' and 'Model'
- ▶ Ensured numeric columns (Price, Mileage) were in correct format



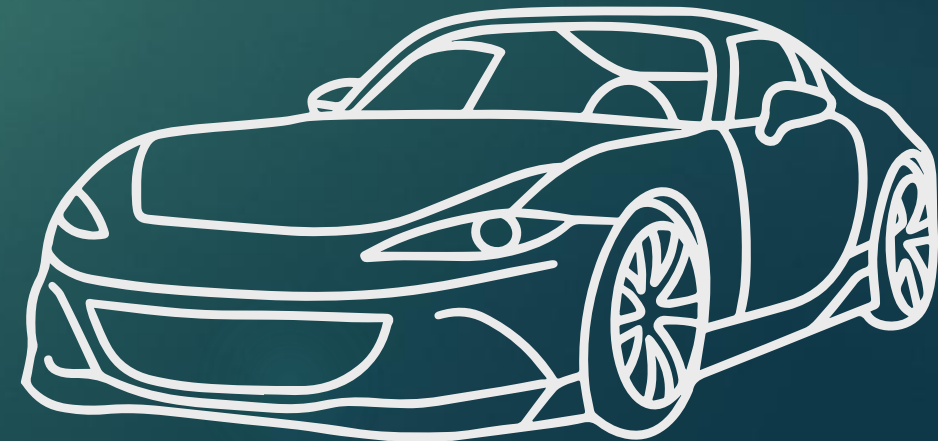
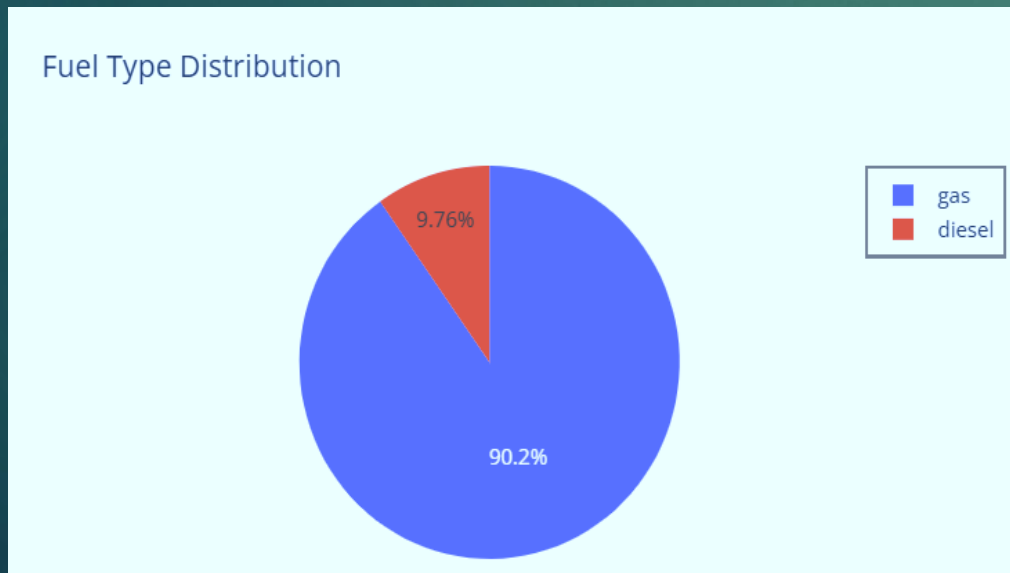
Average price by brand

- ▶ Visual: Bar Chart
- ▶ Shows which brands have higher average prices.
- ▶ The bar chart of average car price by brand reveals significant variation in pricing across different manufacturers.
- ▶ Luxury brands such as Jaguar, Buick, Porsche, and BMW commanded the highest average prices, indicating their focus on premium vehicles.



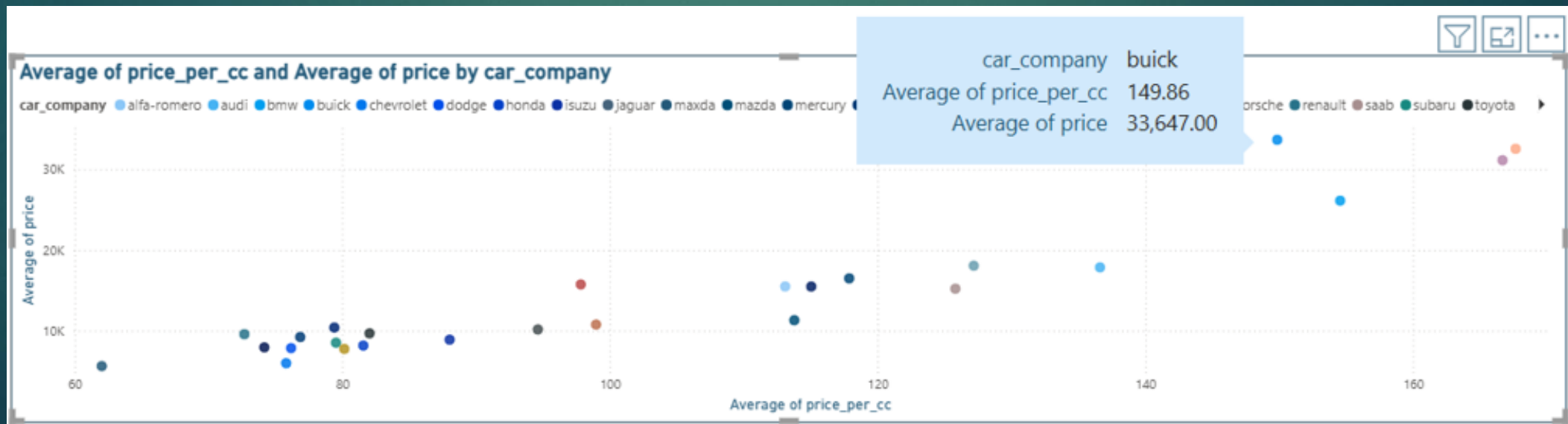
Fuel Type Distribution

- ▶ Visual: Pie or Stacked Column Chart
- ▶ Tools used: Here we used Plotly to create a clean, interactive pie chart that shows the proportion of cars powered by each fuel type (e.g., petrol, diesel).
- ▶ Highlights the distribution of cars by fuel type.
- ▶ Insight: This pie chart visualizes the distribution of different fuel types in the dataset and gas dominates listings, followed by diesel.



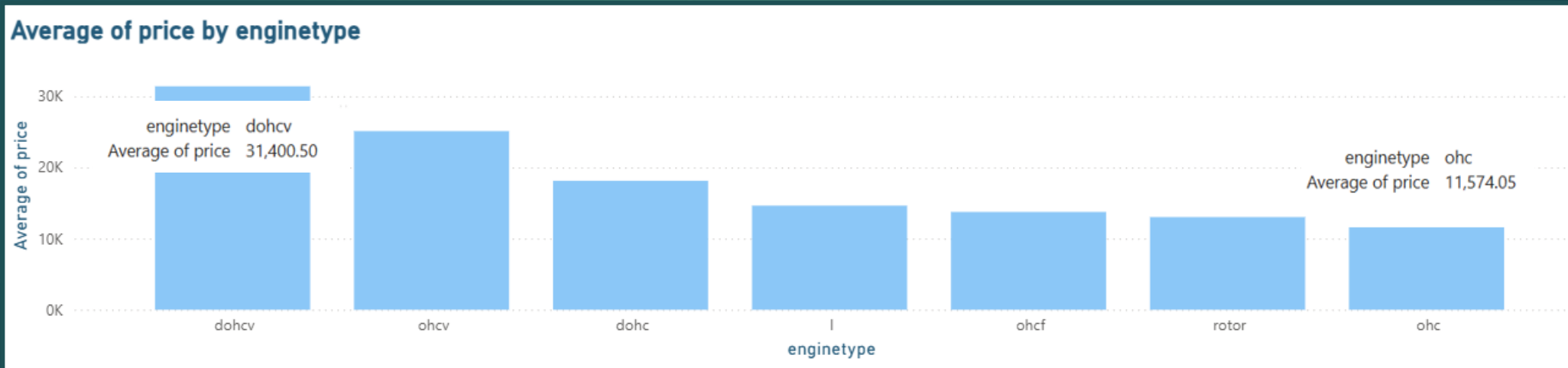
Price per cc vs price by brand

- ▶ Visual: Scatter Chart
- ▶ Displays the relationship between price per cc and price by brand.
- ▶ Insights: Some brands (e.g., Jaguar, Porsche, Buick) are clustered at a higher price and higher price_per_cc, showing the performance/luxury positioning.
- ▶ Other brands show efficiency-focused or budget placement with lower price_per_cc and the average price.



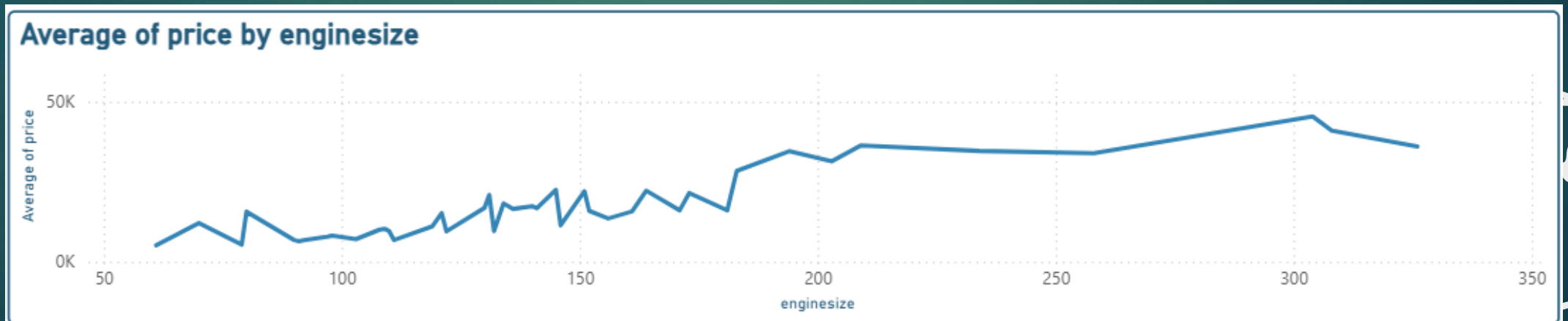
Average Car Price by Engine Type

- ▶ Visual : Column chart
- ▶ Shows the effect of engine type on price.
- ▶ Insight: Engine type significantly impacts vehicle pricing. Higher-performance engines (like dohc, ohcv) are linked with premium vehicles, while ohc engines dominate the budget segment.



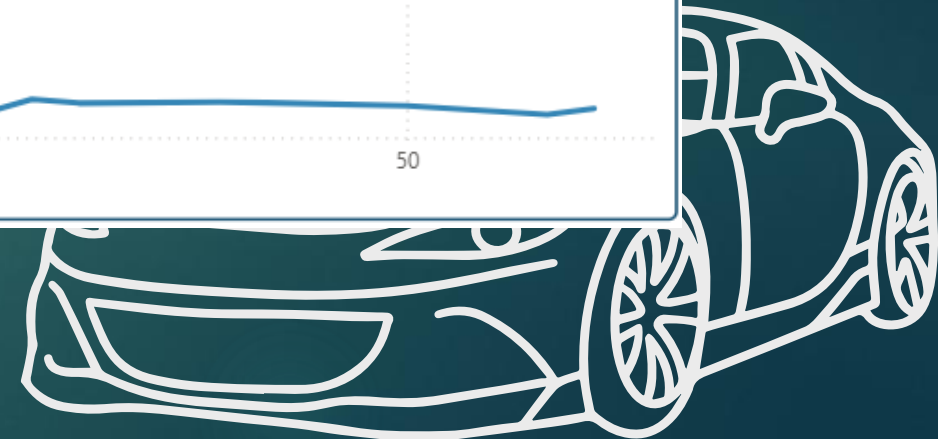
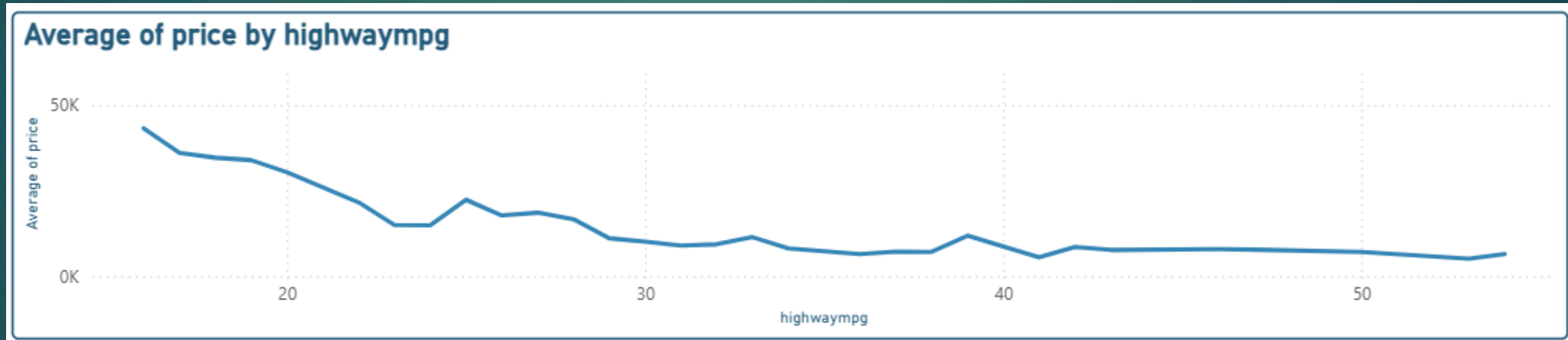
Average Price by Engine size

- ▶ Visual : Line chart
- ▶ There is a strong and clear positive relationship between a car's engine size and its price.
- ▶ Vehicles equipped with larger engines tend to be significantly more expensive.



Average Price by Highway Mileage

- ▶ Visual : Line chart
- ▶ Insights : Highway MPG shows a strong negative correlation with car price, revealing that vehicles with higher fuel efficiency on highways are generally priced lower.



Key Findings

- ▶ Features like engine size, horsepower, and curbweight have a strong positive correlation with price.
- ▶ Cars with more horsepower (and thus better performance) tend to be more expensive. This aligns with consumer expectations in both sports and luxury segments.
- ▶ Fuel efficiency (citympg, highwaympg) have a strong negative correlation suggesting cheaper cars are more fuel-efficient.
- ▶ Some features like compression ratio, stroke, and symboling does not affect the price.



Recommendations

- ▶ Standardize data fields for brand/model to reduce noise.
- ▶ Use power and transmission type in pricing strategy.
- ▶ Develop marketing campaigns focused on premium features (e.g., automatic, high HP).
- ▶ Expand inventory tracking by year and fuel type to forecast future pricing trends.
- ▶ Consider advanced predictive models to estimate car prices based on features.





Thank you

From NaMiPaJay Crew

NA - Nasra Ibrahim

MI - Milos Visnjic

PA - Payal Kanani

JAY - Ajay Chohan